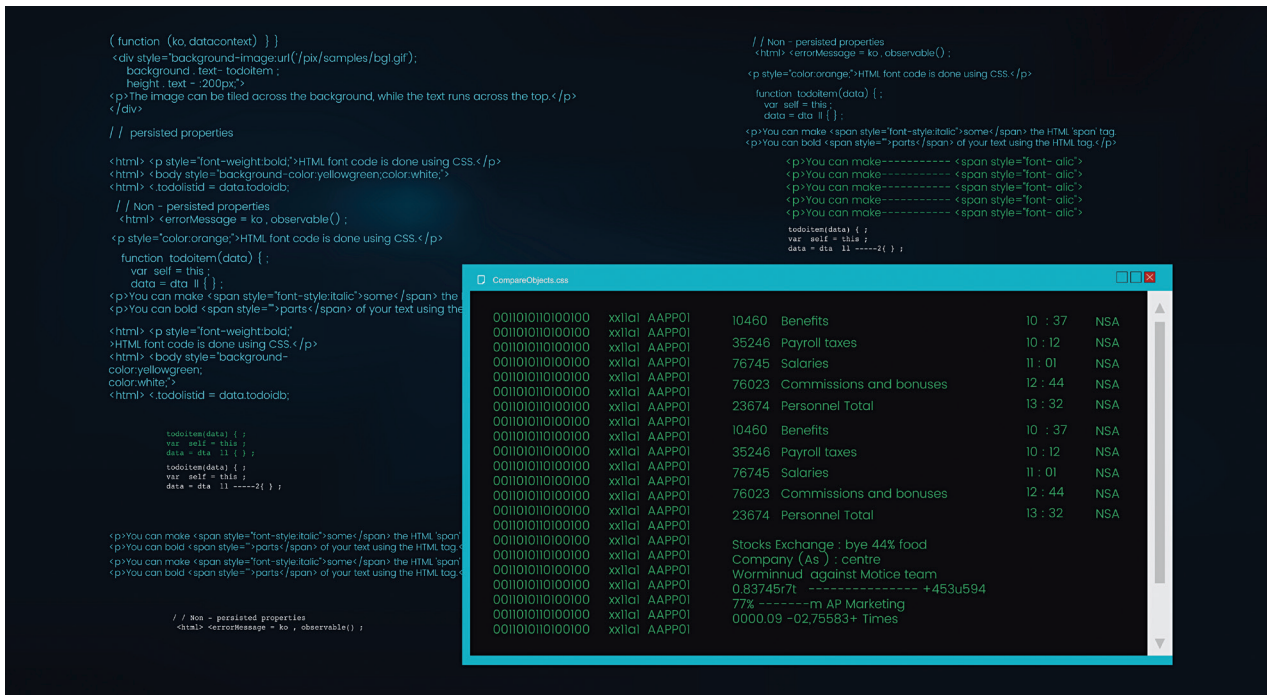




- Enough logic resources to grow into (especially if you plan Chapter 11 capstones)

The two “beginner happy” families

- **ice40 boards:** very approachable, fully open-source flow via Project IceStorm
- **ECP5 boards:** more headroom, open-source flow via Project Trellis + nextpnr-ecp5

If you're buying in India, the exact board model varies by availability. The point isn't the brand; it's the ecosystem support: pinout + open tools + easy programming.

The open-source FPGA flow (the mental map)

Here's the pipeline you'll run again and again:

1. **Write RTL (.v)**
2. **Synthesis (Yosys):** RTL → technology-mapped netlist for that FPGA family
3. **Place & Route (nextpnr):** decide where logic goes and how it routes
4. **Bitstream packer:** generate the actual configuration data for the device (IceStorm/Trellis related tooling)
5. **Upload/program:** send bitstream to FPGA SRAM (volatile) or flash (non-volatile)

Two key tools you'll see in the wild:

Yosys

Yosys is an open-source framework for Verilog RTL synthesis. It also ships packaged “platform synth” commands like `synth_`

`ice40` and `synth_ecp5`, which run target-specific scripts.

nextpnr

nextpnr is a vendor-neutral open-source place-and-route tool. Its own documentation explicitly calls out support for ice40 (Project IceStorm) and ECP5 (Project Trellis) among others.

Toolchain setup Lab

You can build all of these tools from source. You can also spend two weekends doing it and discovering new spiritual dimensions of dependency errors.

For a beginner Fast Track, the cleanest approach is OSS CAD Suite, which is a binary distribution bundling many open-source tools used in digital logic design.

Step-by-step: install OSS CAD Suite

1. Download OSS CAD Suite from its releases page (linked from Yosys docs).
2. Extract it to a simple folder path (avoid spaces if you can).
3. Use the provided environment script to add tools to your PATH for the terminal session. (OSS CAD Suite's repo is explicit that it's a binary distribution and includes multiple tools.)
4. Verify tools are visible:
 - `yosys -V`
 - `nextpnr-ice40 --help` (if you're on ice40 path)
 - `nextpnr-ecp5 --help` (if you're on ECP5 path)
5. If versions print and help text shows, you're ready.

Why this matters: you want your “learning energy” to go into RTL and hardware debugging, not dependency archaeology.